Validation Infrastructure for Environment-Responsive Coarse-Grained Bead Dynamics

VSEPR-SIM Project

March 19, 2026

Abstract

The companion documents establish the anisotropic surface-mapped bead (ASB) and the environment-responsive extension (ERB), introducing an extended bead state $\mathbb{B}_B = \{B, X_B\}$ with fast observables $(\rho_B, C_B, P_{2,B})$ and a slow internal state variable η_B. This document specifies the *testing and validation infrastructure* developed to verify the environment-responsive layer: synthetic scene construction, trajectory-based time evolution, behavioural assertion frameworks, Monte Carlo robustness harnesses, and structured formation studies at scale $N > 200$. The infrastructure is not auxiliary to the model—it is a co-equal component of model development. Key findings include: (i) the seven reported state variables reduce to four independent degrees of freedom, (ii) large-N runs ($N = 64, 125, 216$) are numerically stable and bounded, (iii) edge versus bulk differentiation emerges naturally from the environment descriptors, and (iv) the causal chain geometry $\rightarrow$ observables $\rightarrow$ target $\rightarrow \eta$ is now empirically confirmed.

Contents

1 Introduction and Position in the Document Series **3**
1.1 Why a Separate Document? . . . 3
1.2 Testing Philosophy . . . 3

2 Scene Construction Methodology **4**
2.1 The `SceneBead` Representation . . . 4
2.2 Deterministic Scene Builders . . . 4
2.3 Randomised Scene Generators . . . 4
2.4 Scene Transforms . . . 5
2.5 Scene Validation . . . 5
2.6 Neighbour List Construction . . . 5

3 Variable Hierarchy **6**
3.1 The Seven Reported Variables . . . 6
3.2 Independence Structure . . . 6
3.3 Causal Chain . . . 6

4 Test Suite Architecture **7**
4.1 Suite Overview . . . 7
4.2 Suite #2: Environment Observable Correctness . . . 7
4.3 Track 2: Monte Carlo Robustness . . . 7
4.4 Suite #3: Structured Formation Studies . . . 8

5 Behavioural Assertion Framework **8**
5.1 Trajectory Assertions . . . 9
5.2 Statistical Assertions . . . 9
5.3 Sweep Harness . . . 9

6 Trajectory Runner **10**
6.1 Single-Bead Trajectory . . . 10
6.2 Multi-Bead Pass . . . 10

7 Key Empirical Findings **10**
7.1 Variable Redundancy . . . 10
7.2 Large-N Stability Gate . . . 11
7.3 Edge vs. Bulk Differentiation . . . 11
7.4 Coordination Histogram Structure . . . 11
7.5 Monte Carlo Robustness . . . 11

8 Infrastructure Architecture **12**
8.1 Header Organisation . . . 12
8.2 PRNG Policy . . . 12
8.3 Dependency Structure . . . 13

9 Validation Gates **13**
9.1 Gate 1: Observable Correctness (Suite #2, Group A) . . . 13
9.2 Gate 2: Dynamical Stability (Suite #2, Groups B–D) . . . 13
9.3 Gate 3: Robustness Under Stochastic Input (Track 2) . . . 14
9.4 Gate 4: Large-N Stability (Suite #3, Phase 2b) . . . 14
9.5 Gate 5: Variable Structure (Suite #3, Phase 1) . . . 14
9.6 Suite #4: Dynamic Formation Regimes . . . 14

10 Completed Extensions (Stage 4) **15**
10.1 Formation-Under-Conditions Experiments . . . 16
10.2 Hysteresis and Memory Tests . . . 16
10.3 Time-to-Equilibrium Quantification . . . 16
10.4 Invariance Validation . . . 16

11 Relationship to Companion Documents **17**

12 Conclusion **17**

1 Introduction and Position in the Document Series

This is the fourth document in the coarse-grained methodology series:

Code	Title	Establishes
CGL	Coarse-Grained and Multi-Scale Layer	Architecture, scale transitions, ensembles
ASB	Anisotropic Surface-Mapped Beads	Descriptor, orientation, interaction engine
ERB	Environment-Responsive Bead Dynamics	X_B, fast observables, η_B, kernel modulation
TIF	**Testing Infrastructure (this document)**	**Validation methodology, empirical findings**

1.1 Why a Separate Document?

Environment-responsive dynamics introduces a new category of model risk. The bead state now includes a dynamical variable (η_B) that evolves via a relaxation equation, couples back into the interaction kernels, and is expected to exhibit memory, hysteresis, and path dependence. None of these properties can be verified by unit-testing individual functions in isolation. They require:

1. **Synthetic scene construction:** controlled spatial configurations of beads with known geometry, spacing, and orientation distributions
2. **Trajectory-based verification:** time-evolution of η under static or controlled conditions, with full observable recording
3. **Behavioural assertions:** monotonicity, boundedness, convergence, and sensitivity checks that go beyond equality tests
4. **Statistical robustness:** thousands of randomised trials verifying that the model does not blow up under stochastic input
5. **Large-N stability:** structured runs with $N > 200$ beads demonstrating that collective behaviour is well-posed

Each of these requires infrastructure that is itself a scientific contribution. This document specifies that infrastructure, catalogues the results it has produced, and establishes the validation gates that must be passed before proceeding to higher-level studies.

1.2 Testing Philosophy

The testing infrastructure follows a **state-first, not chemistry-first** philosophy. Rather than constructing atomistic molecular systems, mapping them to beads, and then testing the environment layer, the infrastructure manufactures bead scenes directly—bypassing atomistic chemistry entirely.

This design choice is deliberate:

- It isolates the environment-responsive layer from mapping errors
- It enables deterministic, reproducible test configurations
- It allows systematic exploration of neighbourhood geometries that may not correspond to any real molecule

- It makes tests fast (microseconds, not minutes)

The philosophy is not that atomistic tests are unimportant, but that *model-level* validation must be independent of the pipeline that feeds it.

2 Scene Construction Methodology

A *scene* is a collection of synthetic beads, each carrying a position $\mathbf{R}_B$, an orientation axis $\hat{n}_B$, and an initial internal state $\eta_B^{(0)}$. Scenes are the fundamental test objects.

2.1 The `SceneBead` Representation

Each synthetic bead is a lightweight struct:

$$\text{SceneBead} = \left\{\mathbf{R},\ \hat{n},\ \texttt{has_orientation},\ \eta^{(0)}\right\} \tag{1}$$

This is intentionally minimal. The bead carries no mass, no charge, no descriptor coefficients, and no channel decomposition. It carries only what the environment-responsive layer needs to compute: positions, orientations, and initial state.

2.2 Deterministic Scene Builders

The following canonical scenes provide controlled geometric configurations with exact, reproducible coordinates:

Scene	N	**Geometry**
Isolated	1	Single bead at origin
Pair	2	Two beads along x-axis at separation d
Linear stack	n	n beads along z-axis, uniform spacing
T-shape	3	Center + aligned (z) + perpendicular (x)
Square	4	Four beads at corners of square in xy-plane
Triangle	3	Equilateral triangle in xy-plane
Tetrahedron	4	Regular tetrahedron vertices
Ring	n	n beads on circle in xy-plane
Dense shell	n+1	Fibonacci spiral on sphere + central bead
Cubic lattice	n^3	Perfect cubic grid
Line bundle	n	Centered z-column, all aligned
Layered slab	n^2k	k square planes stacked along z

Each builder is a pure function: given geometric parameters, it returns a deterministic scene. No random state, no seed, no external dependency.

2.3 Randomised Scene Generators

For robustness testing, randomised generators produce controlled distributions of bead positions and orientations. All use a single deterministic PRNG (`Xorshift32`) seeded explicitly, ensuring exact reproducibility.

Generator	Distribution
Random cluster	N beads uniformly in a box, random orientations
Random cloud	Center at origin + $N-1$ scattered in box
Random shell	Center at origin + N on spherical shell
Separated cloud	Rejection-sampled: minimum inter-bead distance guaranteed
Biased stack	Column along z with controlled jitter and alignment bias
Perturbed lattice	Perfect lattice + bounded random displacement
MC scene	`MCSceneConfig`: $n \in [n_{\min}, n_{\max}]$ neighbours in $[r_{\min}, r_{\max}]$ shell

2.4 Scene Transforms

Two rigid transforms are provided for testing invariance properties:

Translation. $\mathbf{R}_B \to \mathbf{R}_B + \Delta\mathbf{R}$ for all beads. Orientations are unaffected. Environment observables must be translation-invariant, so the environment state of a translated scene must be identical to the original.

Rotation. Both positions and orientations are rotated via Rodrigues' formula about an axis $\hat{k}$ by angle θ:

$$\mathbf{v}' = \mathbf{v}\cos\theta + (\hat{k} \times \mathbf{v})\sin\theta + \hat{k}(\hat{k} \cdot \mathbf{v})(1 - \cos\theta) \tag{2}$$

Applied to both $\mathbf{R}_B$ and $\hat{n}_B$. This is a rigid rotation: inter-bead distances and relative orientations are preserved. All environment observables must be rotation-invariant.

2.5 Scene Validation

Before running any test, a scene can be validated for structural sanity:

Check	What it catches
Minimum pair distance	Catastrophic overlaps, degenerate geometry
Duplicate positions	Copy errors, bad generation
Unit normal verification	Non-unit orientations after rotation or perturbation
Centroid	Drift, asymmetric generation
Bounding box	Scale errors, unexpected extent

These checks prevent debugging generated scenes by staring at coordinates.

2.6 Neighbour List Construction

The neighbour list builder returns *all* other beads in the scene as neighbours for a given bead, regardless of distance. The cutoff filtering is performed inside the environment state kernel (`compute_fast_observables`), which applies distance-based Gaussian weighting controlled by r_c and σ_ρ.

This design ensures that scene construction is completely independent of kernel parameters. The same scene can be evaluated under different cutoff regimes without regeneration.

3 Variable Hierarchy

The environment-responsive bead reports seven state variables. A central finding of the testing infrastructure is that these seven variables are *not* seven independent degrees of freedom.

3.1 The Seven Reported Variables

Variable	**Symbol**	**Type**	**Range**
Local density	ρ	Fast observable	$[0, \infty)$
Normalised density	$\hat{\rho}$	Derived	$[0, 1]$
Coordination number	C	Fast observable	$[0, N-1]$
Orientational order	P_2	Fast observable	$[-\frac{1}{2}, 1]$
Normalised order	$\hat{P}_2$	Derived	$[0, 1]$
Target function	f	Composite	$[0, 1]$
Internal state	η	Dynamical variable	$[0, 1]$

3.2 Independence Structure

Systematic sweeps across canonical scenes (Suite #3, Phase 1) reveal:

- $\hat{\rho}$ is a deterministic rescaling of ρ: $\hat{\rho} = \rho/\rho_{\max}$
- $\hat{P}_2$ is a deterministic rescaling of P_2: $\hat{P}_2 = (P_2 + \frac{1}{2})/\frac{3}{2}$
- f is a linear combination of normalised forms: $f = \alpha\hat{\rho} + \beta\hat{P}_2$

Therefore the system has four independent variables:

$$\boxed{\text{Independent:} \quad \rho,\ C,\ P_2,\ \eta \qquad \text{Derived:} \quad \hat{\rho} = g(\rho),\ \hat{P}_2 = h(P_2),\ f = \alpha\, g(\rho) + \beta\, h(P_2)} \tag{3}$$

This is not a deficiency—it is a clarification. The normalised forms and the target function are not independent scientific degrees of freedom. They are intermediate computational quantities.

3.3 Causal Chain

The empirically confirmed causal structure of the environment-responsive layer is:

$$\underbrace{\text{geometry, spacing, orientation}}_{\text{scene configuration}} \longrightarrow \underbrace{\rho,\ C,\ P_2}_{\text{measured local state}} \longrightarrow \underbrace{\hat{\rho},\ \hat{P}_2}_{\text{normalised forms}} \longrightarrow \underbrace{f}_{\text{target}} \longrightarrow \underbrace{\eta}_{\text{memory}} \tag{4}$$

This chain means that η is not a primary descriptor of the scene. It is a *stateful memory variable* that integrates the scene's geometric properties through the intermediate observable layer. If someone asks "what determines η?", the answer is not "the scene directly" but "the scene determines descriptors, descriptors determine the target, the target drives η."

This causal interpretation should govern all subsequent test design, parameter fitting, and scientific reporting.

4 Test Suite Architecture

The validation infrastructure is organised into five test suites, each targeting a different aspect of model behaviour. All suites share a common assertion pattern and report format.

4.1 Suite Overview

Suite	Tests	File	Focus
Suite #1	60	`test_anisotropic_model.cpp`	Channel kernels, per-ℓ decomposition, SH rotation
Suite #2	77	`test_environment_suite2.cpp`	Observable correctness, η dynamics, constitutive response, coupled pipeline, stress/pathology
Track 2	39	`test_track2_robustness.cpp`	Monte Carlo robustness across 4000+ random scenes
Suite #3	52	`test_formation_suite3.cpp`	Structured $N > 10$ formation studies, atlas, large-N
Suite #4	113	`test_formation_regimes_suite4.cpp`	Dynamic regimes, invariance, condition-to-structure
Suite #5	102	`test_ensemble_proxy_suite5.cpp`	Emergent effective medium mapping, ensemble response proxies
CLI Suite	72	`test_cg_cli.cpp`	Integration tests for all CG console commands
Total	**515**		

All 515 tests pass. No test is skipped or expected to fail.

4.2 Suite #2: Environment Observable Correctness

Suite #2 is the most structurally detailed, organised into five groups:

Group A: Environment observable correctness. Neighbourhood sensing, translation invariance, rotation invariance, distance sweeps, isolated bead special cases.

Group B: Internal state dynamics. η evolution under constant environment, τ-dependence, determinism under identical initial conditions, timestep convergence.

Group C: Constitutive response. Kernel modulation factors, backward compatibility (all $\gamma_k = 0$ recovers unmodulated kernels), modifier boundedness.

Group D: Coupled pipeline. Multi-step evolution, history-dependent behaviour, emergent ordering under sustained alignment.

Group E: Stress and pathology. Extreme values ($\eta < 0$, $\eta > 1$), NaN/Inf guards, empty neighbour lists, very large distances, degenerate orientations.

4.3 Track 2: Monte Carlo Robustness

Track 2 runs large ensembles of randomised local bead scenes and verifies statistical properties. It is a property-based statistical validation, not a unit test.

Ensemble	N **trials**
Isotropic random	1000
Aligned dense	1000
Mixed distribution	2000

Properties verified across all 4000 trials:

- $\eta \in [0, 1]$ at all timesteps
- No NaN or Inf in any observable
- Isotropic environments do not spuriously activate ordered state
- Aligned dense environments correctly activate ordering
- Kernel modulations remain finite and sign-correct
- Higher alignment bias produces higher P_2 (monotonic trend)

4.4 Suite #3: Structured Formation Studies

Suite #3 consists of three phases:

Phase 1: Single-variable response atlas. For each of seven environment variables, sweep a controlling scene parameter (spacing, from 2.0 to 15.0) while holding all others fixed. Run across six canonical scenes. Record output range, monotonicity, and perturbation sensitivity.

Phase 2a: Pairwise coupling maps. Sweep two variables simultaneously on coarse 5×5 grids. Four coupling maps: $(\rho \times \eta)$, $(C \times P_2)$, $(P_2 \times \hat{P}_2)$, $(\rho \times f)$. Detect nonlinear transitions, saturation, and divergence.

Phase 2b: Large-N runs. $N = 64$, 125, 216 beads with controlled initial conditions across six scene types: perfect lattice, perturbed lattice, line bundle, layered slab, shell initialisation, random cloud. All 18 runs stable and bounded.

5 Behavioural Assertion Framework

Traditional unit tests assert equality: *expected output equals actual output*. Environment-responsive testing requires *behavioural* assertions that characterise qualitative properties of trajectories and distributions.

5.1 Trajectory Assertions

Assertion	Meaning
`is_monotone_increasing`	Sequence is non-decreasing (within 10^{-15} tolerance)
`is_monotone_decreasing`	Sequence is non-increasing
`is_bounded`(v, a, b)	All values in $[a, b]$
`approaches_value`(v, t, ϵ)	Final value within ϵ of target t
`has_variation`	Not all values are identical (system responded)
`is_finite`	No NaN or Inf anywhere in the sequence

These are not ad hoc checks. Each corresponds to a physical property of the relaxation equation:

- Monotone increasing η when $f > \eta$ (driving toward target from below)
- Bounded in $[0, 1]$ by construction of the clamped relaxation
- Convergence to f at steady state
- Variation confirms the system is not degenerate
- Finiteness guards against numerical pathology

5.2 Statistical Assertions

For ensemble-level verification:

Tool	Purpose
`compute_stats`	Mean, variance, percentiles (5th, 50th, 95th), NaN/Inf counts
`mean_increases_with`(x, y)	Binned monotonic trend: mean of y increases with sorted x
`groups_separated`(A, B, δ)	Mean of group B exceeds mean of group A by at least δ

These tools enable the Track 2 robustness harness to make statistical claims across thousands of trials without hand-inspecting individual runs.

5.3 Sweep Harness

The direct sweep harness automates single-variable parameter studies:

1. Accept a scene-generating function $S(v)$ parameterised by sweep variable v
2. Linearly sample v from $v_{\min}$ to $v_{\max}$
3. For each sample: build scene, evolve all beads for n steps, record bead 0 state
4. Return a vector of `ResponseRecord` with all seven environment variables

This enables systematic parameter studies with a single function call, replacing ad hoc loops in test files.

6 Trajectory Runner

The trajectory runner is the central execution primitive. It time-evolves a single bead within a static (frozen-position) scene and records the full history of all observables.

6.1 Single-Bead Trajectory

Given a scene, a bead index, and initial $\eta^{(0)}$:

1. Build the neighbour list for the target bead (all other beads in the scene, unfiltered)
2. For each timestep $t = 0, 1, \ldots, n-1$:
   (a) Compute fast observables (ρ, C, P_2) from the current neighbour configuration
   (b) Normalise: $\hat{\rho} = \rho/\rho_{\max}$, $\hat{P}_2 = (P_2 + \frac{1}{2})/\frac{3}{2}$
   (c) Compute target: $f = \alpha\hat{\rho} + \beta\hat{P}_2$
   (d) Integrate: $\eta(t{+}1) = f + (\eta(t) - f)\, e^{-\Delta t/\tau}$
   (e) Record: (η, f, ρ, P_2, C) appended to trajectory
3. Return the complete `EtaTrajectory`

Because positions are frozen, the fast observables are constant across timesteps. Only η evolves. This makes the trajectory runner a test of the relaxation dynamics in isolation, cleanly separated from the question of whether positions would change under the modified forces.

6.2 Multi-Bead Pass

For large-N studies, the multi-bead runner evolves all beads simultaneously. At each timestep, every bead's environment state is updated from its own neighbour list. The initial η for each bead is read from the `SceneBead::eta` field, enabling non-uniform initialisation for hysteresis and memory tests.

7 Key Empirical Findings

The testing infrastructure has produced several findings that inform subsequent model development.

7.1 Variable Redundancy

As documented in Section 3, the seven reported state variables contain only four independent degrees of freedom. The normalised variables $\hat{\rho}$ and $\hat{P}_2$ are deterministic rescalings of ρ and P_2 respectively, and the target function f is a linear combination of the normalised forms.

This finding has two consequences:

1. Subsequent analysis should treat ρ, C, P_2, and η as the primary state, not all seven variables
2. If the target function is later generalised to a nonlinear form, the redundancy structure will change and must be re-evaluated

7.2 Large-N Stability Gate

Eighteen structured runs at $N = 64$, 125, and 216 across six scene types (perfect lattice, perturbed lattice, line bundle, layered slab, shell, random cloud) all produced:

- $\eta \in [0, 1]$ for all beads at all timesteps
- No NaN or Inf in any observable
- Bounded, finite output across all configurations

This clears the large-N stability gate. The environment-responsive layer does not exhibit numerical instability under any tested structured initialisation at scale $N \leq 216$.

7.3 Edge vs. Bulk Differentiation

On the $6 \times 6 \times 6 = 216$ cubic lattice:

Property	**Bulk**	**Edge**
ρ	high	low
η	high	low
C	high	low

This differentiation is physically expected: interior beads have more neighbours, higher density, and therefore higher η at steady state. Edge beads have fewer neighbours and lower observables.

The fact that this emerges naturally from the environment descriptors— without explicit surface detection or boundary tagging—validates that the observables carry physically meaningful topological information.

7.4 Coordination Histogram Structure

On a $5 \times 5 \times 5 = 125$ cubic lattice, the coordination number C exhibits a structured distribution corresponding to geometric position:

Position	**Coordination**
Corner beads	lowest C
Edge beads	intermediate C
Face beads	higher C
Interior beads	highest C

The histogram shows at least two distinct coordination values, confirming that the environment descriptors resolve topological position within a structured formation.

7.5 Monte Carlo Robustness

Across 4000 randomised trials:

- No numerical divergence in any trial
- Isotropic random environments produce low η (no spurious activation)
- Aligned dense environments correctly produce high η

- Higher alignment bias monotonically increases P_2
- Dense shells produce higher C than sparse shells

These are not individual test results—they are statistical properties of the model verified across an ensemble.

8 Infrastructure Architecture

8.1 Header Organisation

The testing infrastructure is split into three headers:

Header	Contents
`scene_builders.hpp`	`SceneBead`, `Xorshift32`, all scene builders, `translate_scene`, `rotate_scene`, scene validation helpers, `build_neighbours`, `set_initial_eta`
`test_runners.hpp`	`EtaTrajectory`, `run_trajectory`, `run_all_beads`, behavioural assertions, `StatSummary`, `compute_stats`, `RunRecord`, `run_robustness_trial`, `ResponseRecord`, `run_single_variable_sweep`, `CoordinationHistogram`
`scene_factory.hpp`	Backward-compatible aggregator: includes both of the above

The split separates concerns:

- *What* to test (scene construction) is independent of *how* to test (trajectory execution, assertion checking)
- New scene types can be added without touching the runner
- New assertion helpers can be added without touching the builders
- Existing consumers include `scene_factory.hpp` and automatically receive both

8.2 PRNG Policy

All randomised generation uses a single PRNG implementation (`Xorshift32`) with explicit seeding. This ensures:

- Exact reproducibility: same seed $\Rightarrow$ same scene
- Platform independence: no reliance on `std::mt19937` distribution implementation
- Debuggability: any failing trial can be reproduced by seed
- Determinism: required by the project's anti-black-box philosophy

8.3 Dependency Structure

Header	$\longrightarrow$	Depends on
`scene_builders.hpp`	$\longrightarrow$	`environment_state.hpp`, `state.hpp` (Vec3)
`test_runners.hpp`	$\longrightarrow$	`scene_builders.hpp`, `environment_coupling.hpp`
`scene_factory.hpp`	$\longrightarrow$	Both of the above

The dependency on `environment_state.hpp` is necessary because `build_neighbours` returns `NeighbourInfo` structs, and the runners call `update_environment_state`. The dependency on `environment_coupling.hpp` is needed only by `run_robustness_trial` for kernel modulation reports.

9 Validation Gates

The testing infrastructure defines explicit gates that must be passed before proceeding to higher-level studies.

9.1 Gate 1: Observable Correctness (Suite #2, Group A)

All fast observables must be:

- Translation-invariant
- Rotation-invariant (when both positions and orientations are rotated)
- Monotonic in expected directions (density increases with neighbour count)
- Zero for isolated beads ($\rho = 0$, $C = 0$, P_2 undefined/zero)
- Finite for all non-degenerate inputs

Status: Passed. 77/77 tests in Suite #2.

9.2 Gate 2: Dynamical Stability (Suite #2, Groups B–D)

The slow variable η must:

- Remain in $[0, 1]$ at all times
- Converge to f at steady state
- Exhibit monotone approach when starting below/above target
- Be deterministic under identical initial conditions
- Be insensitive to timestep within reasonable range

Status: Passed. All dynamical tests in Suite #2 Groups B–D.

9.3 Gate 3: Robustness Under Stochastic Input (Track 2)

The model must not produce:

- NaN or Inf for any random scene configuration
- η outside $[0, 1]$
- Spurious ordering in isotropic environments
- Failure to order in aligned dense environments

Status: Passed. 39/39 tests across 4000 random trials.

9.4 Gate 4: Large-N Stability (Suite #3, Phase 2b)

At $N = 64, 125, 216$:

- All beads bounded
- All observables finite
- Edge/bulk differentiation present
- Coordination histogram structured

Status: Passed. 18/18 large-N runs stable.

9.5 Gate 5: Variable Structure (Suite #3, Phase 1)

The response atlas must confirm:

- Identified redundancy pattern ($\hat{\rho}$, $\hat{P}_2$, f are derived)
- Four independent degrees of freedom
- Monotonic or bounded response across all scenes
- Finite output across the full sweep range

Status: Passed. 52/52 tests in Suite #3.

9.6 Suite #4: Dynamic Formation Regimes

Suite #4 (113 tests) implements the dynamic, invariance, and regime-mapping studies described below. It is organised into four phases:

Phase 4A: Dynamic validation (21 tests). Hysteresis under spacing ramps (compress then decompress), alignment ramps (fully aligned vs. random orientation), high-η vs. low-η initialisation comparison, relaxation time extraction, and τ-variation effects. Key findings:

- Compressed and decompressed scenes remain bounded and finite
- Low-η and high-η initialisations converge to the same final mean η (difference < 0.05), confirming the system is a first-order relaxation without multistability
- Relaxation timing $t_{10\%} \leq t_{5\%} \leq t_{1\%}$ with all thresholds reached within the run window
- Faster τ produces faster relaxation, as expected

Phase 4B: Invariance validation (18 tests). Translation invariance, rigid rotation covariance (Rodrigues'), bead-order permutation invariance (Fisher–Yates shuffle), seeded reproducibility, and rotation composition. All passed:

- η, ρ, C, P_2 are exactly invariant under translation
- All observables and η are invariant under rigid rotation (tolerance 10^{-10})
- Sorted per-bead observables are identical under permutation
- Fixed seeds produce bit-identical scenes and trajectories; different seeds produce different scenes
- Two composed 90° rotations match a single 180° rotation

Phase 4C: Formation-regime atlas (29 tests). Sweeps across six scene families (lattice, perturbed lattice, line bundle, layered slab, shell, separated cloud), five α/β weight settings, five τ values, and five alignment bias levels. Bulk/edge classification via coordination threshold. Results:

- All spacing sweeps bounded, finite, and responsive
- α/β weight variation affects η
- All τ sweep runs stable
- Alignment bias measurably changes η and P_2
- Non-degenerate geometries show clear bulk/edge differentiation with bulk $C \geq$ edge C
- Uniform-distance geometries (shell) correctly show degenerate coordination

Phase 4D: Medium-to-large-N scaling (45 tests). $N = 64, 125, 216$ across four scene types (perfect lattice, perturbed lattice, shell, separated cloud). Plus cross-N scaling consistency and formation history convergence:

- All 12 large-N runs bounded and finite
- Bulk interior η consistent across N (max difference < 0.15)
- Late-stage η variance $\leq$ early-stage variance, confirming convergence
- Coordination histograms valid at all sizes

10 Completed Extensions (Stage 4)

The following were planned extensions that have now been implemented and verified in Suite #4.

10.1 Formation-Under-Conditions Experiments

Systematic variation of:

- Density and spacing (compression/expansion)
- Alignment bias (ordered $\leftrightarrow$ disordered)
- Shell count and local coordination
- Target function weights (α, β)
- Relaxation timescale (τ)
- Initial η distributions

Status: Completed. All sweeps bounded, finite, and responsive.

10.2 Hysteresis and Memory Tests

Tested path dependence:

- Compress then decompress spacing
- Aligned vs. random orientations
- Initialise η low vs. high under same scene

Status: Completed. The system converges to the same final state regardless of initial η (no multistability detected), confirming pure first-order relaxation dynamics.

10.3 Time-to-Equilibrium Quantification

Measured:

- Time to within 1%, 5%, 10% of final value
- Correct ordering: $t_{10\%} \leq t_{5\%} \leq t_{1\%}$
- τ-dependent relaxation speed
- Convergence of late-stage variance (monotone relaxation)

Status: Completed. No oscillation, overshoot, or numerical fragility detected across all tested regimes.

10.4 Invariance Validation

Newly established invariance proofs:

- Translation invariance (exact)
- Rigid rotation covariance (tolerance 10^{-10})
- Bead-order permutation invariance (sorted observables)
- Seeded reproducibility (bit-identical)

Status: Completed. All invariances confirmed.

11 Emergent Effective Medium Mapping (Stage 5)

The ensemble proxy layer bridges per-bead environment state to macroscopic response characterisation. It does not claim to predict continuum-scale constitutive properties; it provides *proxy fields* computed from the distributions and spatial organisation of bead-state variables.

11.1 Architectural Position

Environment-state evaluation (per-bead)
↓
Ensemble statistics / spatial field summaries
↓
Macroscopic response proxies
↓
Later constitutive or transport models

The analysis module lives at `coarse_grain/analysis/ensemble_proxy.hpp` and produces an `EnsembleProxySummary` from a vector of `EnvironmentState` and bead positions.

11.2 Summary Sections

The `EnsembleProxySummary` struct contains seven sections:

[label=()]**First moments:** mean ρ, $\hat{\rho}$, C, P_2, $\hat{P}_2$, η, f, and mean $|\eta - f|$ **Second moments:** $\mathrm{var}(\rho)$, $\mathrm{var}(C)$, $\mathrm{var}(P_2)$, $\mathrm{var}(\eta)$, $\mathrm{var}(f)$ **Edge/bulk contrasts:** bulk and edge bead counts, gap in ρ, C, P_2, η **Occupancy fractions:** fraction of beads above coordination, η, and alignment thresholds **Spatial autocorrelation:** neighbour-pair autocorrelation of η, ρ, P_2 **Response proxies:** five scalar proxies bounded in $[0, 1]$ **Diagnostics:** all-finite and all-bounded flags

11.3 Response Proxy Definitions

Five macroscopic response proxies are defined:

Cohesion $(\bar{\hat{\rho}} + \bar{\eta} + (1 - \overline{|\eta - f|}))/3$. Higher for dense, stabilised, low-mismatch systems.

Uniformity $1 - (d_\eta + d_\rho)/2$ where $d = \mathrm{clamp}(2\sqrt{\mathrm{var}}, 0, 1)$. Higher for homogeneous systems.

Texture $\overline{\hat{P}_2}$. Higher for oriented, anisotropic organisation.

Stabilization $\bar{\eta}\,(1 - \overline{|\eta - f|})$. Higher for converged, well-adapted states.

Surface sensitivity Mean fractional $|\text{bulk} - \text{edge}|$ gaps in ρ, C, η. Higher when bulk and edge environments are sharply distinct.

11.4 Suite #5: Validation (102 tests)

Suite #5 validates the ensemble proxy layer across five phases:

Phase	Focus	Tests
5A	Computation correctness (means, variances, bounds)	30
5B	Physical sense (ordering, response, coherence)	14
5C	Large-N stability ($N = 64, 125, 216$)	32
5D	Invariance (translation, rotation, permutation, seed)	16
5E	Sensitivity (sweeps, families, parameter response)	10
Total		**102**

Key verified properties:

- All five proxies remain in $[0, 1]$ across all tested scenes and parameters
- Dense lattices produce higher cohesion than sparse ones
- Aligned stacks produce higher texture than random clouds
- Converged systems have higher stabilisation than fresh ones
- All proxies are translation, rotation, and permutation invariant
- Results are seeded-reproducible (bit-identical)
- Large-N proxies remain finite, bounded, and consistent across $N = 64 \to 216$

Companion document	This document
ERB defines $X_B = \{\rho, C, P_2, \eta\}$	Verifies X_B correctness, stability, invariance
ERB specifies relaxation dynamics	Tests convergence, boundedness, monotonicity
ERB predicts emergent behaviour	Observes edge/bulk differentiation, coordination structure
ERB introduces six new parameters	Sweeps parameters, confirms no blow-up
ERB claims hysteresis and metastability	Suite #4 confirms relaxation, no multistability
ERB + TIF establish variable hierarchy	Suite #5 constructs ensemble proxies from those variables

The relationship is explicit: ERB makes theoretical claims, TIF provides the machinery to test them, and the test results either confirm the claims or reveal where the theory requires amendment.

12 Conclusion

The testing infrastructure for environment-responsive coarse-grained bead dynamics is not a peripheral convenience—it is a co-equal scientific instrument. It has:

1. Established 515 tests across seven suites, all passing
2. Identified the variable hierarchy: four independent degrees of freedom (ρ, C, P_2, η), three derived quantities ($\hat{\rho}$, $\hat{P}_2$, f)
3. Confirmed the causal chain: geometry $\to$ observables $\to$ target $\to$ η
4. Cleared the large-N stability gate at $N = 216$
5. Demonstrated edge/bulk differentiation and coordination structure

6. Verified robustness across 4000+ randomised trials
7. Confirmed translation, rotation, and permutation invariance
8. Completed formation-under-conditions, hysteresis, relaxation timing, and regime-atlas experiments (Suite #4)
9. Established that the system is a pure first-order relaxation with no multistability or path-dependent memory
10. Constructed and validated five macroscopic response proxies (cohesion, uniformity, texture, stabilization, surface sensitivity) from bead-state ensemble statistics (Suite #5)
11. Provided the validated foundation for Emergent Effective Medium Mapping

The model is not merely defined. It is tested, measured, and characterised. The testing infrastructure ensures that subsequent development—including constitutive property determination—is grounded in empirical evidence rather than theoretical assumption.